



SERANGOON JUNIOR COLLEGE
General Certificate of Education Advanced Level
Higher 1

CANDIDATE
NAME

CLASS

INDEX NUMBER

CHEMISTRY
Preliminary Examination
Paper 2

8872/02
17 August 2012
2 hours

Additional Materials: Data Booklet
 Writing Papers

Candidates answer Section A on the Question Paper.

READ THESE INSTRUCTIONS FIRST

Write your name and class on all the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use a soft pencil for any diagrams, graphs or rough work.

SECTION A:

Answer **all** questions in the space provided.

SECTION B:

Answer any **two** questions on separate answer paper.

At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [] at the end of each question or part question.

For Examiner's Use	
MCQ P1	
Section A	
B5	
B6	
B7	
Total	

This document consist of **14** printed pages and **1** blank page

Section A

For
Examiner's
Use

Answer **all** the questions in this section in the spaces provided.

- 1 (a)** Aluminium is the third most abundant element (after oxygen and silicon) found in the Earth's crust. Due to its strong affinity to oxygen, it is almost never found in the elemental state. Aluminium reacts readily with carbon to form aluminium carbide, Al_4C_3 , with relative formula mass of 144.0. It is stable up to 1400 °C and it decomposes in water to produce methane.

- (i)** Define the term *relative formula mass*.

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- (ii)** State the electronic configuration of aluminium in aluminium carbide.

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- (iii)** Draw the dot-and-cross diagram to illustrate the bonding in aluminium carbide.

- (iv)** Calculate the percentage by mass of carbon in aluminum carbide.

[4]

(b) Aluminium carbide can react with protic reagents such as hydrochloric acid to produce methane gas and aluminium(III) chloride. At high temperatures, the molecular mass of aluminium(III) chloride is half that measured under room conditions.

- (i)** State the molecular formula of aluminium(III) chloride at room temperature and suggest a reason for the difference in molecular mass of aluminium(III) chloride at high temperature and room temperature.

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- (ii)** Write a balanced equation for the reaction between aluminium carbide and hydrochloric acid.

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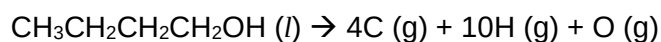
- (iii)** Using the equation in **(b) (ii)**, calculate the maximum volume of methane gas that can be produced when 12.0 g of aluminium carbide reacts with 250 cm³ of 2.0 mol dm⁻³ hydrochloric acid at standard temperature and pressure.

[6]

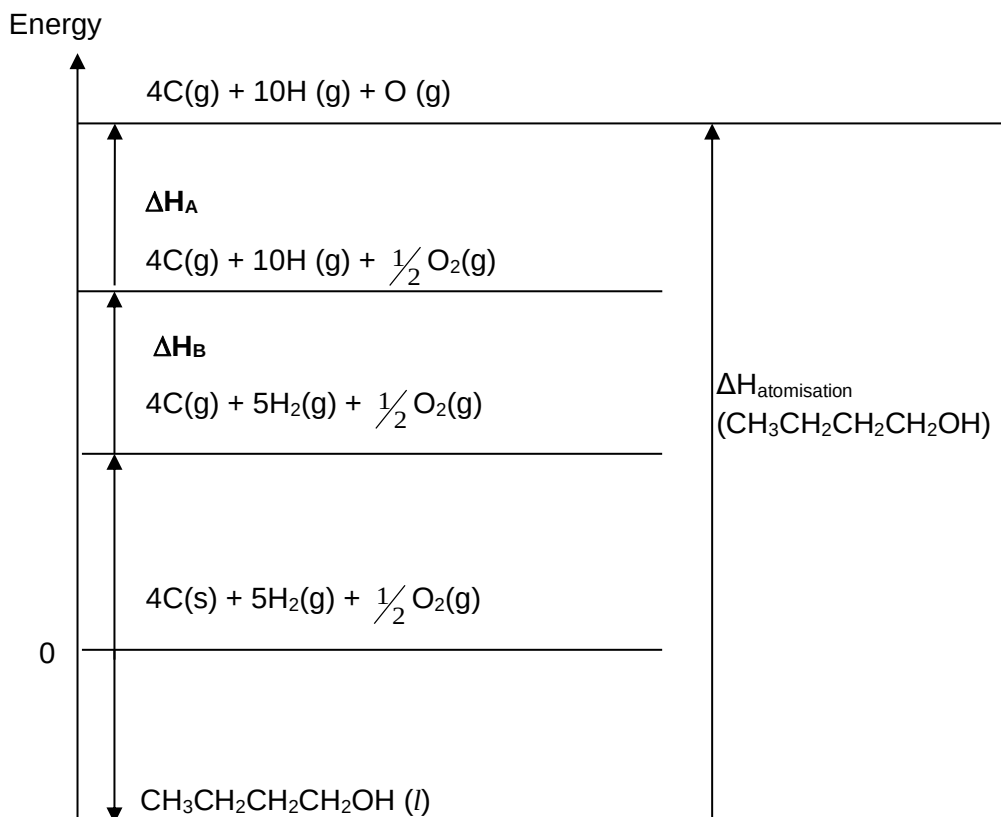
Total: [10]

- 2 (a) The standard enthalpy change of atomisation of a molecule is the energy absorbed when one mole of a gaseous molecule is atomised to form the gaseous atoms under standard conditions.

The atomisation reaction of butan-1-ol is:



An energy level diagram involving the atomisation of butan-1-ol is shown below:



- (i) Define the term *standard enthalpy change of formation*.

.....

.....

.....

- (ii) Use the *Data Booklet* to obtain the values of the enthalpy change for ΔH_A and ΔH_B .

ΔH_A :

ΔH_B :

- (iii) Hence, using the information below and the values from (a) (ii), calculate the enthalpy change of atomisation of butan-1-ol.

Given:

$$\Delta H_{\text{atomisation}}(\text{C}) = 716 \text{ kJ mol}^{-1}$$

$$\Delta H_{\text{formation}}(\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{OH}) = -328 \text{ kJ mol}^{-1}$$

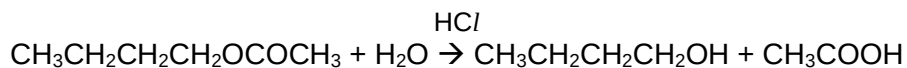
- (iv) Calculate another value for the enthalpy change of atomisation of butan-1-ol using **only bond energies** values quoted from the *Data Booklet*.

- (v) Bond energies quoted from the Data Booklet are average values and may result in disparity in the answers in (a) (iii) and (a) (iv). Suggest another reason for the difference in the values calculated in (a) (iii) and (a) (iv).

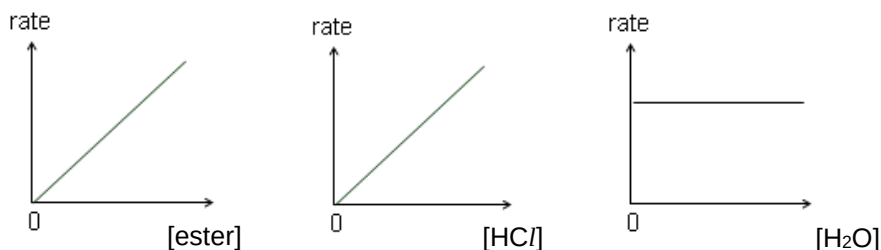
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[6]

- (b) Butan-1-ol can be produced by the acidic hydrolysis of butyl ethanoate.



The following graphs were obtained when the rate and concentration of the reactants were investigated in the laboratory.



From the graphs, deduce the rate equation and hence determine the units for the rate constant.

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 [2]

- (c) The school laboratory has run out of alkaline iodine. Propose a series of reaction to differentiate bottles containing pentan-1-ol and pentan-2-ol.

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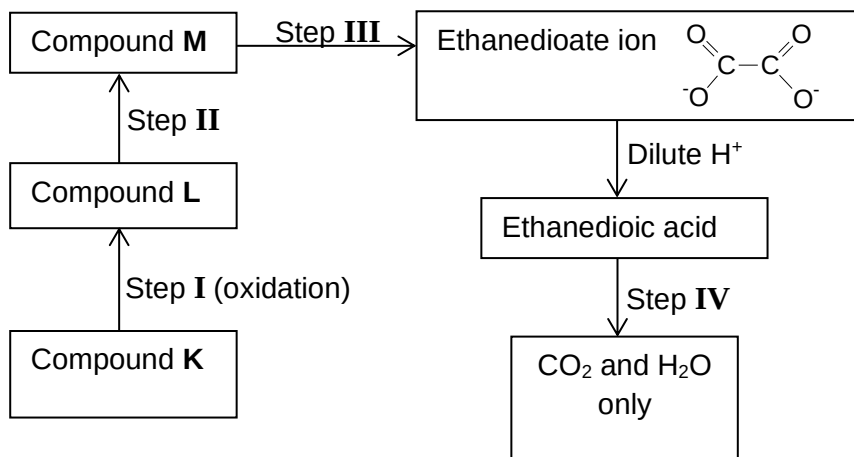
 [2]

Total: [10]

- 3 (a) The reaction scheme below shows how compound **K** can be converted into ethanedioic acid.

Compound **K** has molecular formula, C_4H_8O . It forms a yellow precipitate of CHI_3 when reacted with warm alkaline aqueous iodine. Compound **K** is also known to decolourise bromine water.

Compound **L** and **M** react with warm alkaline aqueous iodine to form a yellow precipitate. **M** evolves white fumes when reacted with phosphorus pentachloride but does not react with carbonate. On the other hand, only compound **L** decolourises bromine water.



- (i) From the reaction scheme and the information provided, draw the displayed formulae of compounds **K** and **M**.

Compound **K**:

Compound **M**:

- (ii) Suggest why potassium manganate(VII) cannot be used for step I.

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- (iii) State the reagents and conditions for steps **II**, **III** and **IV**.

Step **II**:

Step **III**

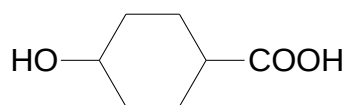
Step **IV**:

[6]

- (b) (i) Propose how ethanoic acid can be converted into ethanal.

- (ii) Hot ethanoic acid followed by ethanol was reacted with compound **T** in the presence of concentrated H_2SO_4 .

Draw the structural formula of the organic product formed.



Compound **T**

- (iii) There is an isomer of compound **T** which has a lower boiling point. Draw this isomer.

Total: [10]

- 4 The following table shows some properties of period 3 elements **V**, **W**, **X**, **Y** and **Z**.

Element	Boiling point / °C	Electrical conductivity	Reaction with oxygen
V	1410	0.10	No reaction
X	649	0.42	Burns very vigorously and brightly to form oxide
W	660	1.00	Solid formed reacts with acid and bases
Y	119	0	Forms a gaseous product
Z	44	0	Burns vigorously in excess oxygen

- (a) Oxide of **X** and oxide of **Z** are dissolved in 2 separate beakers. The content in the beakers are then mixed. State the type of reaction that will occur. Explain your answer

.....

[2]

- (b) Explain the difference in boiling points of **V**, **X** and **Y**, with reference to their bonding and structure.

.....

[3]

- (c) Write two chemical equations to show how the oxides of **X** and **W** reacts with hydrochloric acid.

.....
.....

[2]

- (d) Describe the reactions, if any, of the chlorides of element **W** with water, suggesting the pH of the resulting solution and writing equations, where appropriate.

.....
.....
.....
.....

[3]

Total: [10]

Section B

Answer **two** questions from this section on separate answer paper.

- 5(a)** 15 cm³ of hydrocarbon **J** was exploded in excess oxygen and allowed to cool to room temperature. There was an overall reduction of 45 cm³. After the gas mixture was passed through aqueous sodium hydroxide a further contraction of 120 cm³ was observed.

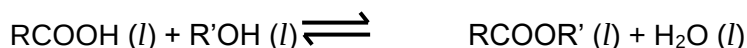
J reacts with gaseous hydrogen bromide to form **K**. Aqueous sodium hydroxide was added to **K** to produce **L**.

L decolourises aqueous potassium dichromate(VI) to give **M**. Both **L** and **M** give a yellow precipitate with aqueous alkaline iodine. **K** reacts with aqueous sodium cyanide to produce **N**, which forms **P** upon reaction with aqueous hydrochloric acid.

- (i) Using the combustion data given, determine the molecular formula of **J**.
- (ii) Using your answer in (a) (i) deduce, with reasoning, the structures of compounds **J**, **L**, **M** and **P**.

[12]

- (b)** Esters can be synthesized from the reaction of a carboxylic acid and an alcohol. The equation for the reaction is shown below:

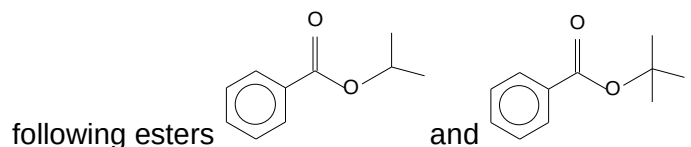


In an experiment, 1 dm³ of 2.5 mol dm⁻³ of acid is reacted with 1 dm³ of 1.5 mol dm⁻³ of alcohol. The equilibrium constant, K_c of this reaction is 1.6.

- (i) Define *dynamic equilibrium*.
- (ii) Write an expression for the equilibrium constant, K_c . Calculate the concentration of the ester formed at equilibrium.

[4]

- (c)** Esters are commonly used in food flavouring. Describe a chemical test to distinguish the



[2]

- (d)** On reaction with ethanolic silver nitrate, bromocyclohexane forms more precipitate as compared to chlorocyclohexane while no precipitation was observed for fluorocyclohexane. Explain why these observations were made.

[2]

Total: [20]

- 6(a)** Most of the Earth's crust consists of solid oxides. They were formed when elements develop an oxide coating as a result of oxidation in air. Aluminium foil, often used in food preparation, develops a thin oxide layer over the aluminium metal to protect it from corrosion.

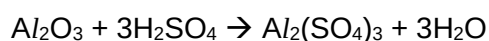
- (i) The oxides of magnesium, silicon and phosphorus differ in their reactions with water.

Account, with reference to the structure and bonding of the oxides, for the above reactions and state the approximate pH of the solutions formed. Write equations where appropriate.

- (ii) A student attempted to remove the oxide layer on the aluminium foil.

He placed the aluminium foil strips into a polystyrene cup and added 60 cm³ of 1.2 mol dm⁻³ sulfuric acid.

3.0 g of the aluminium oxide coating the metal reacted based on the following reaction.



The maximum temperature change was recorded to be 11.5 °C.

Given that the specific heat capacity of the solution = 4.2 J g⁻¹ K⁻¹ and assuming the process to be 75% efficient, calculate the standard enthalpy change of neutralisation for the reaction.

- (iii) Another student proposed a method to remove the aluminium oxide layer using aqueous sodium hydroxide instead. With the aid of balanced chemical equation(s), comment on whether you agree with this proposal.
- (iv) With reference to your answer in (a)(ii), predict if the standard enthalpy change of neutralisation between ethanoic acid and aluminium oxide is less exothermic or more exothermic.

[12]

- (b)** Compound **W** is found to contain 75.3% carbon, 10.6% hydrogen and 14.1% oxygen by mass. The relative molecular mass of **W** is 112.

When **W** is oxidised with acidified potassium manganate(VII), a compound **X**, C₄H₆O₃, is produced together with propanoic acid. No white fumes are evolved when **W** was mixed with phosphorus pentachloride.

X produces a pale yellow precipitate when warmed with alkaline solution of iodine, accompanied by a solution which, when treated with dilute hydrochloric acid, gave compound **Y**.

Y may also be obtained when NC – CH₂ – CN is heated with a dilute acid.

On heating with concentrated sulphuric acid, 2 mole of **X** reacted with 1 mole of ethane-1,2-diol to produce a sweet smelling compound **Z**.

Deduce the structure of compounds **W**, **X**, **Y** and **Z** and explain the chemistry of the reactions described above.

[8]

Total: [20]

- 7 An acidic buffer consists of a *weak acid*, such as a carboxylic acid, and the salt of its conjugate base. Within a certain pH range, the acidity of a carboxylic acid is important and affects the buffering capacity of the buffer solution. The acid dissociation constant of the carboxylic acid is directly proportional to its buffer capacity.

- (a) Define the term *weak acid*. [1]
- (b) 25.0 cm³ of citric acid, HA, with a pH of 3.1 undergoes neutralisation with 25.60 cm³ of 0.50 mol dm⁻³ of potassium hydroxide. Support your answer with relevant argument why citric acid is a weak acid. [2]
- (c) With the aid of equations, explain how citric acid and its sodium salt function as a buffer solution. [2]
- (d) Rank the following carboxylic acids based on increasing pH:
2-chloropropanoic acid, 2-fluoropropanoic acid, 2-methylpropanoic acid
Explain your answer. [4]
- (e) An unknown ester produces 2-methylpropanoic acid as the only product when it was subjected to the following reactions:
1. Heating with dilute sulfuric acid followed by
 2. Heating with aqueous potassium manganate(VII)

In an investigation, the following kinetics data were obtained for Step 1.

With 1.0 mol dm⁻³ sulfuric acid:

Time / min	0	2	4	6	8	10
[ester] / mol dm ⁻³	0.0100	0.0068	0.0047	0.0032	0.0021	0.0015

With 2.0 mol dm⁻³ sulfuric acid:

Time / min	0	2	4	6	8	10
[ester] / mol dm ⁻³	0.0100	0.0046	0.0021	0.0010	0.0004	0.0002

- (i) Draw the structural formula of the ester.
- (ii) Using the same axes, plot graphs of [ester] against time for:
- (a) Reaction with 1 mol dm⁻³ of sulfuric acid.
 - (b) Reaction with 2 mol dm⁻³ of sulfuric acid.
- Hence, find the rate equation showing your working clearly.
- (iii) In the reverse reaction, where esterification takes place between 2-methylpropanoic acid and an alcohol, the addition of a small amount of concentrated sulfuric acid increases the rate of reaction significantly. With the help of suitable diagram, explain how concentrated sulfuric acid affects the rate of reaction.

Suggest if the addition of concentrated sulfuric acid affects the equilibrium constant.

[11]
Total: [20]

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